

NOTES FOR MATH 3150, WEEK 2

1. SOLVING $x^2 = 2$.

We cannot solve many algebraic equations in $\mathbb{Q}$. For example, we have the following:

Proposition 1.1. *The equation $x^2 = 2$ has no rational solutions.*

Proof. Assume by contradiction that there exists a rational number

$$x = \frac{p}{q}$$

so that

$$x^2 = 2.$$

Of course, we may assume that p and q have no common divisors, $(p, q) = 1$.

We have that

$$\left(\frac{p}{q}\right)^2 = 2,$$

so

$$p^2 = 2q^2.$$

If p were an odd number, say $p = 2k + 1$, then

$$p^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

so p^2 is odd as well, because

$$p^2 = 2m + 1 \quad \text{for } m = 2k^2 + 2k.$$

That is impossible, because $p^2 = 2q^2$, so p^2 is even.

In conclusion, p is an even number, say $p = 2k$. Then

$$2q^2 = p^2 = 4k^2,$$

hence $q^2 = 2k^2$, which implies as above that $q = 2h$.

But that contradicts $(p, q) = 1$. This means that $x^2 = 2$ has no rational solution. $\square$

Remark 1.2. *By Pythagoras' theorem we know that the diagonal of a square of side equal to one must have length equal to $\sqrt{2}$. The previous proposition says that we must enlarge the set of rational numbers to solve $x^2 = 2$.*

2. THE REAL NUMBERS $\mathbb{R}$

We define real numbers abstractly by specifying their properties first. We will assume $\mathbb{R}$ is a set of numbers that contains $\mathbb{Q}$ and satisfies the same set of axioms A, M, DL and O as the rational numbers. We will define an important additional axiom that distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

Definition 2.1. Let S be a nonempty subset of $\mathbb{R}$.

- (1) We say S is bounded above if there exists $M \in \mathbb{R}$ so that $x \leq M$, for all $x \in S$. In this case, M is called **an upper bound** of S .
- (2) We say S is bounded below if there exists $m \in \mathbb{R}$ so that $x \geq m$, for all $x \in S$. In this case, m is called **a lower bound** for S .
- (3) If S has both an upper and a lower bound, we say that S is bounded.
- (4) If a set S admits an upper bound $M \in S$, then we say M is **the maximum** of S . Similarly, if $m \in S$ is a lower bound for S , we say m is **the minimum** of S .

Let us practice with a few examples.

Examples.

- (1) The interval

$$S = [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

is bounded, $m = a$ is a lower bound for S , and $M = b$ is an upper bound for S . In fact, according to the above definition, $M = b$ is the maximum of S and $m = a$ is the minimum of S .

Let us note that $\tilde{M} = b + 10$ is also an upper bound for S .

- (2) The interval

$$T = (a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$

is bounded, $m = a$ is a lower bound and $M = b$ is an upper bound.

In fact, as in the previous example, $M = b$ is the maximum of T because b is an upper bound for T and $b \in T$.

However, T does not have a minimum! Indeed, suppose that $m \in T$ were the minimum of T . Since $a < m \leq b$, it follows that $\bar{m} = \frac{m+a}{2}$ also satisfies $a < \bar{m} \leq b$. Since $\bar{m} < m$, this is a contradiction to m being the minimum. Note that any element in the interval (a, m) will provide this contradiction.

However, $m = a$ still is quite special for T , it is what we call the infimum of T .

Definition 2.2. Let S be a nonempty subset of $\mathbb{R}$.

1. We say $M \in \mathbb{R}$ is **the supremum** of S , and denote $M = \sup S$, if M is an upper bound for S and no number $\bar{M} < M$ is an upper bound for S .

2. We say $m \in \mathbb{R}$ is **the infimum** of S , and denote $m = \inf S$, if m is a lower bound for S and no number $\bar{m} > m$ is a lower bound for S .

Let us note that if S has a maximum M , then $\sup S = M$.

Example. The interval

$$S = (a, b) = \{x \in \mathbb{R} : a < x < b\}$$

has $\sup S = b$ and $\inf S = a$. None of them are achieved, so S does not have either maximum nor minimum.

Let us see why $\sup S = b$:

Indeed, b is an upper bound for S because any $x \in S$ satisfies $x \leq b$ by definition of S . Suppose by contradiction that $\sup S < b$. Then, denoting $M = \sup S$, we have that $M < b$, so $M \in S$. This means that M is the maximum of S . We follow the argument in Example (2) above to reach a contradiction. Indeed, we then have that $\frac{M+b}{2} \in S$ and $M < \frac{M+b}{2}$, contradicting that M is an upper bound for S . In conclusion, $\sup S = b$.

2.1. The Completeness axiom. We now arrive at the first very important concept in analysis.

Axiom 2.3 (The Completeness Axiom). *Every nonempty subset S of $\mathbb{R}$ that is bounded above has a supremum.*

We do not need to specify a similar property for infimum, that follows from the sup axiom. Let us discuss now an important consequence.

The Archimedean Property: If $a, b > 0$, there exists $n \in \mathbb{N}$ so that $na > b$.

Proof. We prove that for any real number $x > 0$, there exists $n \in \mathbb{N}$ so that $n > x$.

Then the Archimedean Property follows applying this for $x = \frac{b}{a}$.

Assume by contradiction that the above is not true, so $n \leq x$ for all $n \in \mathbb{N}$. This implies that $\mathbb{N}$ is bounded above, so it admits a supremum.

Let $M = \sup \mathbb{N}$. Then $M - 1$ is not an upper bound for $\mathbb{N}$, therefore there exists $n \in \mathbb{N}$ so that

$$n > M - 1.$$

Therefore, $M < n + 1$, which contradicts M being an upper bound for $\mathbb{N}$. $\square$

2.2. Further properties. Assume that A, B are bounded above.

- (1) If $a \leq b$ for all $a \in A$ and $b \in B$, then $\sup A \leq \sup B$.
- (2) $M = \sup A$ if and only if M is an upper bound of A and for any $\varepsilon > 0$, there exists $a \in A$ so that $M \leq a + \varepsilon$.
- (3) For any $a > 0$, there exists $n \in \mathbb{N}$ so that $a > \frac{1}{n}$.
- (4) If x is an arbitrary real number, prove that there exists exactly one integer n which satisfies $n \leq x < n + 1$. This n is called the greatest integer in x and is denoted with $[x]$.
- (5) For any $x, y \in \mathbb{R}$, $x < y$, there exists $z = \frac{p}{q} \in \mathbb{Q}$ so that $x < z < y$.

Proof. We prove (2), (4), and (5), and leave the rest to the reader.

We first prove (2). For the direct statement, for any $\varepsilon > 0$ we have that $\sup A - \varepsilon < \sup A$. Since $\sup A$ is the least upper bound of A , this implies that $\sup A - \varepsilon$ is not an upper bound of A , because otherwise it would be a smaller upper bound than $\sup A$. Therefore, there exists $a \in A$ such that $\sup A - \varepsilon < a$. This proves that $\sup A \leq a + \varepsilon$ as claimed.

Let us prove the converse now. That is, assume that M is an upper bound for A and that for any $\varepsilon > 0$, there exists $a \in A$ so that $M \leq a + \varepsilon$. We aim to show that $M = \sup A$. Since M is an upper bound of A , it follows that $\sup A \leq M$, because $\sup A$ is the least upper bound of A . On the other hand, for any $\varepsilon > 0$ there exists $a \in A$ such that $M - \varepsilon \leq a$. Since $a \leq \sup A$, this proves that $M - \varepsilon \leq \sup A$. But $\varepsilon > 0$ is arbitrary, which shows that $M \leq \sup A$ as otherwise, if $M > \sup A$, we could pick $\varepsilon = \frac{M - \sup A}{2} > 0$ contradicting that $M - \varepsilon \leq a$. In conclusion, $M = \sup A$.

We now prove (4). We show such an integer exists. Let's fix $x \in \mathbb{R}$, and consider the set

$$A := \{n \mid n \in \mathbb{Z} \text{ and } n \leq x\}.$$

This set is nonempty, because $\mathbb{Z}$ is unbounded below. The set A is evidently bounded above by x , therefore it admits a supremum, by the sup axiom. Let

$$a = \sup A.$$

Since a is the least upper bound of A , and x is an upper bound of A , we know that $a \leq x$. Furthermore, we claim that a is an integer i.e. $a = \max A$.

Indeed, notice that $a - \frac{1}{2}$ cannot be an upper bound of A , so there exists $n \in A$ so that

$$a - \frac{1}{2} < n.$$

The claim that a is an integer will follow if we can show that $a = n$. In this sense, we check that n is an upper bound for A .

Suppose this is not true, then there exists $m \in A$, so that $m > n$. But because both m and n are integers, it means that $m \geq n + 1$.

This contradicts the fact that

$$m \leq a < n + \frac{1}{2}.$$

This shows that indeed n is an upper bound for A , hence it is the maximum of A and $a = n$.

So we have an integer n with the property that $n \leq x$. If now $n + 1 \leq x$, that would mean that $n + 1 \in A$, and contradict the maximality of n . So, we have

$$x < n + 1.$$

This proves the existence of n as required by the problem.

Now the uniqueness of this n . Suppose that there exist two integers n and m so that

$$n \leq x < n + 1 \text{ and } m \leq x < m + 1.$$

This implies that $n < m + 1$ and $m < n + 1$. Hence,

$$-1 < n - m < 1,$$

and the only integer strictly between -1 and 1 is zero, therefore $n - m = 0$, thus showing $n = m$.

We now prove (5). It suffices to consider $y > x > 0$. We know there exists $n \in \mathbb{N}$ so that

$$y - x > \frac{1}{n},$$

so

$$ny > nx + 1.$$

Let

$$m = [nx] + 1,$$

so $m \in \mathbb{N}$.

By 4. we know that

$$nx < m \leq nx + 1,$$

therefore

$$nx < m < ny.$$

This implies that

$$x < \frac{m}{n} < y$$

as claimed. □

Let us see how property (2) above can be applied to prove that

$$\sup S = b \quad \text{for } S = (a, b).$$

Clearly, b is an upper bound for S . For any $\varepsilon > 0$ let $x = b - \varepsilon$. We may assume that ε is small enough so that $x > a$, so that $x \in S$. By our choice of x we certainly have that $b \leq x + \varepsilon$. According to property (2), this means that $b = \sup S$.

2.3. The equation $x^2 = 2$ revisited. We now establish the existence of $\sqrt{2}$.

Theorem 2.4. *There exists a positive real number $x > 0$ so that $x^2 = 2$.*

Proof. Define

$$S = \{x \in \mathbb{R} : x^2 \leq 2 \text{ and } x > 0\}.$$

Clearly, S is nonempty and bounded above by $x = 2$. Let

$$M = \sup S.$$

We want to prove that

$$M^2 = 2.$$

If this is not true, then

$$\text{either } M^2 > 2 \text{ or } M^2 < 2.$$

Assume first that $M^2 < 2$, so $M \in S$. By the properties proved above, there exists $n \in \mathbb{N}$ so that

$$2 - M^2 > \frac{1}{n}.$$

Now for

$$x = M + \frac{1}{10n}$$

we get

$$\begin{aligned} x^2 &= M^2 + \frac{M}{5n} + \frac{1}{100n^2} \\ &< 2 - \frac{1}{n} + \frac{2}{5n} + \frac{1}{100n^2} \\ &< 2. \end{aligned}$$

This means

$$M + \frac{1}{10n} \in S,$$

therefore

$$M + \frac{1}{10n} \leq M,$$

which is a contradiction.

Now let us assume that $M^2 > 2$. Then there exists $n \in \mathbb{N}$ so that $M^2 - 2 > \frac{1}{n}$. Since $M = \sup S$, there exists $a \in S$ so that

$$a \leq M < a + \frac{1}{10n}.$$

It follows that

$$\begin{aligned} 2 + \frac{1}{n} &< M^2 \\ &< a^2 + \frac{a}{5n} + \frac{1}{100n^2} \\ &< 2 + \frac{2}{5n} + \frac{1}{100n^2}, \end{aligned}$$

which is again a contradiction.

In the last line above we used that $a \in S$, so $a^2 \leq 2$, and that 2 is an upper bound for S .

In conclusion, $M^2 = 2$.

□